

COMPONENTS

DC-DC converter

MORNSUN has released an upgraded 5-200W ultra-wide input voltage range DC-DC converter series that can deliver better performance and cost-effectiveness for applications involving photovoltaic combiner boxes, inverters, energy storage BMS, charging piles, and more.



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The PV15-27BxxR3 series has an optional input voltage range of 100V to 1,000V DC. This series has been designed to provide input under-voltage protection, meet UL1741 standard, and be suitable for high-altitude harsh outdoor environments.

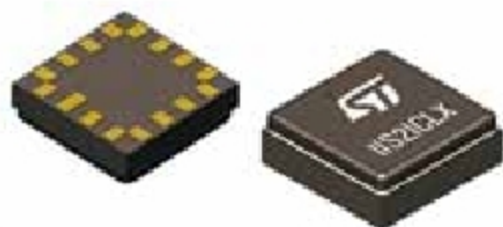
These modules feature an isolation voltage of 4000V AC, output over-voltage protection, short circuit protection, and input against reverse protection, which ensures circuit stability.

MORNSUN

www.mornsun-power.com

Inclinometer

STMicroelectronics has launched IIS2ICLX, a high-accuracy, low-power, 2-axis digital inclinometer for use in applications such as industrial automation and structural-health monitoring. It features a programmable machine-learning core and sixteen independent programmable finite state machines that help edge devices save power and reduce data transfers to the cloud.



With an advanced embedded function, the IIS2ICLX lowers system-level power consumption and extends the operation of battery-powered nodes.

The sensor's inherent characteristics simplify integration into high-performing products while minimising the effort and cost of calibration.

The IIS2ICLX is housed in a high-performance ceramic-cavity LGA package measuring 5mm × 5mm × 1.7mm.

STMicroelectronics

www.st.com

MEMS microphone

TDK Corporation has introduced InvenSense ICS-40638 MEMS analogue microphone, ideal for wearable and IoT applications, particularly in outdoor, industrial, or harsh environments.



The ICS-40638 includes a MEMS microphone element, an impedance converter, and a differential output amplifier.

The InvenSense ICS-40638 is now available in a small 3.5mm × 2.65mm × 0.98mm bottom port surface-mount package from authorised distributors.

TDK

www.tdk.com

Motor control IC

Renesas Electronics Corporation has introduced a 48V mobility winning combination solution that helps customers accelerate the development of e-scooters, e-bikes, UPS, and energy storage systems.



It has a modular approach towards hardware and software that can be implemented for many 24V to 48V applications such as lawnmowers, electric carts, robot cleaners, power tools, power banks, and more.

The solution also uses fifteen Renesas ICs, including three key devices—ISL94216 16-cell battery front end (BFE), HIP2211 100V MOSFET drivers, and RX23T 32-bit microcontroller (MCU)—for motor control.

The 48V mobility winning combination solution includes two boards that address a mobility application's high battery cell count and power levels.

Renesas Electronics

www.renesas.com

Photorelays

Toshiba Electronic Devices & Storage Corporation has launched three photorelays—TLP3407SRA, TLP3475SRHA, and TLP3412SRHA. These are small voltage-driven devices that can be mounted in high-temperature areas, making it easier to secure the required temperature design margins for equipment.



The applications of the photorelays include semiconductor testers (memory, SoC and LSI, etc), probe cards, burn-in equipment, and measuring instruments (oscilloscope, data logger, etc). The photorelays' S-VSON4T package has a small mounting area of 2.9mm², which contributes to a reduced PCB size and an increase in the number of photorelays in an existing layout.

The devices are available in the S-VSON4T package with dimensions 2.0mm × 1.45mm × 1.3 mm.

Toshiba

www.toshiba.semicon-storage.com

Wireless charging IC

Renesas Electronics Corporation has introduced its newest member of the wireless power solutions portfolio, the P9415-R wireless power receiver, which is equipped with Renesas' WattShare technology.

The new 15W wireless power receiver